

Fact Sheet: Neurosurgical Capacity in Low- and Middle-Income Countries

The Workforce Gap

There are approximately 73,000 neurosurgeons worldwide, but they are profoundly unevenly distributed. The global median density is 0.44 neurosurgeons per 100,000 people, but this ranges from 0.12 in low-income countries to 2.44 in high-income countries — a 20-fold disparity. Twenty-nine countries have no neurosurgeons at all.

Every year, an estimated 13.8 million people require neurosurgical intervention. More than 5 million essential neurosurgical cases, nearly all in LMICs, go untreated each year.

Pakistan: A Case Study

Pakistan has a neurosurgeon density of 0.14 per 100,000 - roughly one neurosurgeon for every 714,000 people. The country requires four times its current neurosurgical workforce to meet minimum standards set by the World Federation of Neurosurgical Societies (1 per 200,000).

The distribution is severely skewed: 44% of all neurosurgeons in Pakistan practice in just two cities (Karachi and Lahore). TBI management is offered at 65 centers nationwide, but nearly all are concentrated in urban districts. Rural populations - who bear a disproportionate burden of road traffic injuries - have minimal access to neurosurgical care.

Pakistan has approximately 38 practicing rehabilitation medicine consultants, roughly 1,000 physical therapists, and 150 occupational therapists for a population exceeding 230 million. Disability carries significant cultural stigma, and rehabilitation services are virtually nonexistent outside major urban centers.

Infrastructure

Among LMIC institutions reviewed in a multinational study, only 41% reported having neurosurgical capacity. Among those with operative capability, the median neurosurgical intervention rate was just 28%. The median institutional mortality rate for TBI was 19%.

Implementation of electronic medical records and standardized clinical protocols was significantly associated with increased neurosurgical capacity and reduced post-TBI mortality, suggesting that even basic systems-level interventions can improve outcomes.

Access

In South Asia, approximately 53% of the population can reach neurosurgical services within a 2-hour travel window. In Sub-Saharan Africa, that figure drops to 25%. For rural populations in mountainous or geographically isolated regions, access may be effectively zero.

Rehabilitation

Rehabilitation services for brain injury survivors are scarce or nonexistent in most LMICs. In lower-income countries, services such as occupational therapy, speech therapy, cognitive rehabilitation, and vocational retraining are significantly less available than in upper-middle-income countries. In one South African audit, only 2.4% of patients admitted with TBI were discharged to an inpatient rehabilitation facility. Barriers include transport costs, lack of family support, poor understanding of discharge instructions, and absence of rehabilitation professionals in rural areas.

Key Takeaway

The neurosurgical care gap is not just about surgeons: it is a systems failure spanning workforce, infrastructure, equipment, protocols, and post-acute care. Closing this gap requires innovation at every level: training, technology, task-shifting, telemedicine, rehabilitation delivery, and health system design.